

SYSTEMATIC LIQUIDITY RESEARCH

A systematic liquidity-research operation

Cross-instrument evidence from a fifty-instrument, four-hour cross-section.

ABSTRACT

LIQUIDEX is a systematic research operation built on a single principle: institutional liquidity leaves the same structural footprint wherever it trades. To test that principle, one rule-based engine was run across fifty liquid instruments at a single four-hour timeframe, and each instrument was read down to the institutional behaviour that dominates its order flow, under one disciplined per-instrument optimisation protocol. After that read, mean profit factor rose from 0.94 to 1.76 — a 1.87× lift — and 47 of the 50 instruments were profitable, 29 of them crossing from a losing baseline into profit. All figures herein are in-sample and net of commission and slippage; their out-of-sample validation is the subject of Section 5. The breadth of the result is not a feature of the product. It is the evidence for the thesis.

1.87× Mean lift over the raw engine	47 / 50 Instruments profitable once read	4.84 Peak profit factor (SMH)	6.9% Cleanest maximum drawdown (GLD)
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SECTION 1

The mechanism

Institutional order flow is not noise; it is the residue of decisions. A liquidation cascade, a manufactured stop-run, iceberg absorption defending a level, the exhaustion of a one-sided move — each is a behavioural signature, and each repeats.

The engine identifies the signature, attributes it to the institutional behaviour producing it, and resolves that read into a single rule-based position: entry, a structural stop, a target at the opposing wall, a breakeven and a time-stop. It does not merely detect an anomaly; it profiles the actor behind it — what they are doing, where they are going, and whether they are masking it.

An edge that lives in one market is a curve-fit. An edge that recurs across many is a discovery.

The claim under test is narrow and falsifiable. If the edge were specific to one instrument, it would be an artefact of that instrument's history — an overfit. If instead the same behavioural reads recover an edge across many unrelated instruments, the engine is measuring a structural property of institutional markets rather than the idiosyncrasy of a single chart.

That distinction is the entire subject of this document. The sections that follow set out, in order: the protocol used to make the test (Section 2); the results across the full fifty-instrument cross-section (Section 3); two worked examples of the per-instrument read, including the behavioural reasoning behind each (Section 4); and the validation work that separates these in-sample results from a track record (Section 5).

SECTION 2

The research protocol

Breadth is meaningless without method. Every instrument in the cross-section was taken through the same six-step protocol. The discipline is deliberately conservative: a behavioural read is admitted only when it improves the risk-adjusted result at its tuned setting, and discarded the moment it does not.

01 Baseline

The engine is reduced to its core — every optional read disabled, default parameters throughout. Raw profit factor, drawdown and trade count are recorded as the benchmark to beat.

02 Greedy forward-selection

Behavioural reads — cascade-stop, opposing-wall target, the directional-risk gate, risk-velocity confirmation, iceberg absorption, liquidity-sweep exhaustion — are added one at a time. A read is retained only if it improves the risk-adjusted result, and reverted otherwise.

03 Coupled parameter tuning

Each retained read is judged at its tuned setting, not its default: its sensitivity is swept in both directions, and a read that cannot pay at any setting is removed rather than left at a default that flatters it.

04 Two-pass refinement

The surviving configuration is walked a second time, each secondary parameter nudged and immediately reverted if the equity curve degrades. The accepted recipe is the one no further change improves.

05 Risk-adjusted scoring

Every decision is taken on $S = PF / (1 + \text{MaxDD})$, never on profit factor alone, subject to a hard floor of thirty trades. A high profit factor on a small sample does not qualify.

06 Out-of-sample gate

A recipe this specific must clear a held-out window before it is treated as real. In-sample strength is the entry condition for that test, not a substitute for it.

We do not search for the setting that makes a chart look good. We search for the read that survives being doubted.

SECTION 3

Results: the cross-section

The full four-hour cross-section, ranked by final profit factor. **Base** is the raw engine; **Final** is after the protocol of Section 2; **Lift** is their ratio. Two names since taken through the full deep micro-tune (Section 4) are marked in the final column; the remainder are greedy-pass results at default parameters. Three names[†] did not clear the greedy bar.

TABLE 1 — PER-INSTRUMENT RESULTS, 4-HOUR CROSS-SECTION (1-32 OF 50)

#	SYMBOL	CLASS	FAMILY	BASE	FINAL PF	LIFT	MAX DD	TRADES	MICRO-TUNED
1	SMH	ETF	Fade	1.253	4.843	3.87×	21.3%	55	—
2	AMD	Semis	Continuation	1.178	3.354	2.85×	35.6%	74	—
3	LRCX	Semis	Fade	0.892	3.053	3.42×	28.3%	61	—
4	XLK	ETF	Fade	1.595	2.539	1.59×	14.7%	109	—
5	TSEM	Semis	Fade	1.042	2.538	2.44×	24.0%	36	—
6	SOXX	ETF	Fade	1.191	2.493	2.09×	21.3%	84	—
7	GLD	ETF	Fade	0.832	2.457	2.95×	6.9%	43	—
8	XLV	ETF	Continuation	1.098	2.450	2.23×	7.1%	67	—
9	TSM	Semis	Continuation	1.317	2.306	1.75×	29.1%	92	—
10	XOM	Single-name	Fade	1.061	2.284	2.15×	15.1%	51	—
11	QQQ	ETF	Fade	0.936	2.277	2.43×	9.7%	51	—
12	IGV	ETF	Continuation	0.762	2.166	2.84×	16.1%	104	—
13	SPY	ETF	Fade	0.810	2.022	2.50×	9.9%	91	—
14	DIA	ETF	Fade	0.919	1.902	2.07×	12.1%	44	—
15	VOO	ETF	Fade	0.963	1.874	1.95×	7.0%	31	—
16	ARM	Semis	Fade	1.546	1.845	1.19×	20.5%	62	—
17	NVDA	Semis	Continuation	1.283	1.822	1.42×	18.0%	77	—
18	KLAC	Semis	Continuation	1.145	1.769	1.54×	22.4%	121	—
19	VGT	ETF	Fade	0.813	1.767	2.17×	21.1%	75	—
20	SCHD	ETF	Fade	0.858	1.748	2.04×	11.9%	37	—
21	ASML	Semis	Fade	1.191	1.722	1.45×	24.6%	53	—
22	ADI	Semis	Fade	0.692	1.710	2.47×	18.9%	47	—
23	JPM	Single-name	Fade	1.053	1.709	1.62×	29.9%	61	—
24	AMAT	Semis	Fade	0.775	1.668	2.15×	38.3%	67	—
25	XLI	ETF	Continuation	0.999	1.645	1.65×	13.7%	127	—
26	MA	Single-name	Fade	1.101	1.641	1.49×	9.2%	42	—
27	MPWR	Semis	Fade	0.606	1.484	2.45×	24.9%	43	—
28	AVGO	Semis	Continuation	0.846	1.484	1.75×	36.6%	77	—
29	ETN	Single-name	Fade	0.873	1.471	1.68×	17.3%	81	—
30	XLE	ETF	Fade	1.167	1.466	1.26×	18.8%	63	—
31	MCHP	Semis	Continuation	1.023	1.453	1.42×	26.6%	167	—
32	INTC	Semis	Fade	0.839	1.424	1.70×	37.5%	93	—

SECTION 3 (CONTINUED)

Results, continued

TABLE 1 (CONTINUED) – PER-INSTRUMENT RESULTS (33-50 OF 50)

#	SYMBOL	CLASS	FAMILY	BASE	FINAL PF	LIFT	MAX DD	TRADES	MICRO-TUNED
33	TXN	Semis	Fade	0.880	1.409	1.60×	27.0%	94	–
34	COST	Single-name	Fade	0.797	1.389	1.74×	16.7%	91	–
35	MU	Semis	Continuation	0.692	1.385	2.00×	24.9%	108	–
36	DE	Single-name	Fade	1.040	1.374	1.32×	25.5%	92	–
37	ON	Semis	Fade	0.905	1.368	1.51×	49.4%	92	–
38	V	Single-name	Fade	0.858	1.363	1.59×	18.2%	107	–
39	IWM	ETF	Continuation	0.693	1.319	1.90×	22.2%	87	–
40	QCOM	Semis	Continuation	0.858	1.305	1.52×	37.9%	116	–
41	PANW	Single-name	Continuation	0.493	1.250	2.54×	37.5%	155	–
42	ORCL	Single-name	Continuation	0.886	1.220	1.38×	43.0%	128	–
43	XLP	ETF	Continuation	0.731	1.186	1.62×	8.5%	118	–
44	KO	Single-name	Fade	0.956	1.170	1.22×	12.2%	57	–
45	XLY	ETF	Fade	0.790	1.156	1.46×	20.3%	78	–
46	MRVL	Semis	Fade	1.011	1.134	1.12×	43.6%	212	–
47	NXPI	Semis	Fade	0.523	1.035	1.98×	30.4%	57	–
48	XLF [†]	ETF	Continuation	0.923	0.979	1.06×	34.1%	313	–
49	PEP [†]	Single-name	Fade	0.749	0.849	1.13×	28.7%	165	✓ 1.595
50	TLT [†]	ETF	Fade	0.653	0.809	1.24×	26.4%	93	✓ 1.232

[†] XLF, TLT and PEP did not reach a profit factor of 1.0 under the greedy pass; PEP and TLT were subsequently rescued by deep tuning (Section 4), XLF not yet attempted. **Micro-tuned:** ✓ = full deep tune verified live (deep PF shown); – = greedy pass only. All figures in-sample, net of commission and slippage.

What the cross-section shows

Read blind, the cross-section sorts itself into two structural temperaments. The engine resolves **34 of 50 names as faders** – defensives, macro and the broad index complex, where the edge is reverting an exhausted move – and **16 as continuation** names, where the edge is riding the break. **44 of 50 run long-only**, the signature of a passive complex accumulating the book as one coordinated actor. None of this was imposed; it was recovered from price.

BY BEHAVIOURAL FAMILY

FAMILY	NAMES	MEAN PF	MEAN DD
Fade	34	1.79	21.8%
Continuation	16	1.69	25.8%

BY INSTRUMENT CLASS

CLASS	PROFIT.	MEAN PF	BEST
Index & sector ETFs	17 / 19	1.95	4.84
Semiconductors	20 / 20	1.76	3.35
Single-name leaders	10 / 11	1.43	2.28

The read pays most where the flow is most forced: mean profit factor rose from **0.94** to **1.76**,

SECTION 4

Worked examples: the per-instrument read

Two of the three names that failed the greedy pass were not dead — they were misread. Each is opened up below. Together they make the same point from opposite sides: the read is not a setting but an understanding of the actor's temperament, expressed as a configuration.

PEP — *when the desk is patient*

A liquidation cascade never comes for a consumer staple. PepsiCo is not where leverage blows up; it is where patient, mandate-driven capital accumulates and defends. Its flow is quiet, two-sided and mean-reverting — and that is precisely why the default read failed it, listening for violence in a market that trades on patience.

Reading the desk meant matching the algorithm to the psychology: both directions enabled, the reward-to-risk gate raised so only the cleanest reversals qualify, iceberg absorption set deliberately slow to wait for genuinely defended size, and the risk gate tightened on the short side. The coin-flip became a clean, low-drawdown edge.

PEP · PROGRESSION

Raw engine	0.749
Greedy pass	0.849
Deep micro-tune	1.595
Max drawdown	6.5%
Trades	107

Configuration: 11 reads, 6 tuned parameters. Long + short, raised R:R gate, iceberg at 5 hits, high-volume sweep exhaustion, short-side risk gate tightened. Verified live.

TLT — *when the move never reaches the wall*

TLT is long-duration Treasuries — a macro vehicle, not a tape that liquidity desks push to clean technical walls. Its price is dragged by rate expectations and central-bank language, and the full move to the opposing wall rarely completes, because a data print interrupts it first. Everything one adds to it is fragile: tighten the stop and the chop removes you; widen the target and the move dies before it arrives.

What rescued it was not another signal but profit-taking discipline. Banking **60% of the position two-thirds of the way to the wall** — and running a small remainder — stops demanding a completion the macro will not give. The whole result balances on those two Partial-TP values.

TLT · PROGRESSION

Raw engine	0.653
Greedy pass	0.809
Deep micro-tune	1.232
Max drawdown	17.7%
Trades	215

The lever: Partial-TP — bank 60% at 0.66 of the distance to the wall, run a sliver. Long + short, cascade-fade only, wide sweep stop, wall-target. Verified live.

A recipe this specific is where overfitting hides. Both cells carry the out-of-sample flag of Section 5: an in-sample rescue is the entry condition for validation, never a deployment claim.

SECTION 5

Limitations and forward work

The figures in this document are in-sample. The programme below is the standard by which the research is held — the work that separates a strong backtest from a track record. The same discipline is applied to the research itself.

THE VALIDATION STANDARD

TEST	STATUS	NOTE
Commission & slippage	Modelled	Applied to every backtest; all figures are net of costs, not gross.
Capacity	Structural	The book is liquid large-caps and major ETFs that absorb size; the precise ceiling is not yet formally measured.
Out-of-sample	Pending — next	A held-out window the engine has never seen; the immediate priority, and the gate every deep-tuned recipe waits behind.
Walk-forward	Planned	Rolling re-optimisation and re-test, to show each per-symbol recipe is stable through time rather than fitted to one window.
Monte Carlo	Planned	Trade-order and return resampling, to bound the distribution of outcomes and the risk of ruin under sizing.
Live record	The bar	A forward, time-stamped track record — the only dispositive evidence for a fund or licensing conversation.

Two of the three names that failed the greedy pass were rescued only by deep, instrument-specific tuning. Such recipes carry the greatest overfitting risk, and are explicitly withheld from any deployment claim until they clear the out-of-sample gate. The honest summary of the present state is this: the evidence is strong enough to justify that work, and not yet a substitute for it.

SECTION 6

The system, and access

The research is delivered as three instruments, invite-only on TradingView. One engine decides the trade; two companions make its ranking and its risk legible.

LIQUIDEX
THE ENGINE

5,000+ lines of Pine v6. Reads liquidation cascades, order flow, order blocks, iceberg walls and liquidity sweeps, and resolves them into one position — entry, structural stop, and a target at the opposing wall.

Keeper
LIVE RANKING

Runs the engine across the whole book and ranks every name live by profit factor, momentum and risk — so the operator sees the leader the engine prefers, not a guess.

Prophet
RISK GATE

Makes directional risk visible — a composite score, safe and danger zones, and a live position-size verdict that trims exposure as risk rises and stands the operator down in the danger zone.

ACCESS & PARTNERSHIP

Operators receive invite-only access on TradingView: send a TradingView username, access is granted, and the signals arrive as alerts on phone, desktop or a connected trading bot. Access and pricing are available on request.

The defensible asset is the method — a protocol for reading the institutional desks behind the order flow that reproduces across a fifty-instrument cross-section, equities and macro alike. It is open to licensing, a strategic research partnership, or full acquisition.

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